

Grade 10 Maths Unit 4

Unit Outcomes

By the end of this unit, students will be able to:

- ✓ Define the basic trigonometric functions
- ✓ Sketch graphs of basic trigonometric functions
- ✓ Define reciprocals of basic trigonometric functions
- ✓ Identify trigonometric identities
- ✓ Solve some examples on real life problems involving trigonometric equations

4.1 Radian measure of angle: Conversion between radian and degree

Competencies

- ✓ describe radian measure of an angle.
- ✓ convert radian measure to degree measure and vice versa.

4.2 Basic Trigonometric Functions

4.2.1 The Sine, Cosine and Tangent Functions

Competencies

- ✓ use the trigonometric ratios to solve right angled triangles.
- ✓ find the trigonometric values of angles from trigonometric table.
- ✓ find the angle whose trigonometric value is given (using trigonometric table).

4.2.2 Trigonometric Values of Angles

Competencies

- ✓ find the trigonometric values of angles from trigonometric table.

- ✓ find the angle whose trigonometric value is given (using trigonometric table).

4.3 Trigonometric Identities and Equations (10 periods)

Competencies

- ✓ Verify the fundamental trigonometric identities.
- ✓ Simplify trigonometric expressions using algebra and the identities.
- ✓ Solve linear trigonometric equations in sine and cosine.
- ✓ Solve equations involving a single trigonometric function.

4.4 Application

Competencies

- ✓ By using Trigonometric functions to solve real life problems.